UTKARSH DEEP TIWARI

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
Kendriya Vidyalaya No 1 Ordnance Factory, Katni, Madhya Pradesh	Class XII (CBSE)	2014	82.20%
Sacred Heart English Medium Higher Secondary School, Katni, Madhya Pradesh	Class X (CBSE)	2012	8.00%

PROJECTS

Project A: Wind Farm Performance Assessment

Position: PhD Scholar

- Conducted large eddy simulations (LES) to analyze turbulent wake characteristics in wind turbine arrays.
- Developed analytical models to predict wake effects and power output variations in different wind farm configurations.
- Applied machine learning algorithms to reconstruct wind turbine wakes.
- Validated models against field data and benchmarked computational results for robustness.

Project B: Two-dimensional Film Boiling

Position: MTech scholar

- Simulated tandem elliptical surfaces in vertical cross-flow to study flow and heat transfer.
- Numerically investigated film boiling on a rotating cylinder in horizontal cross-flow.

RESEARCH PUBLICATIONS

- Utkarsh Deep Tiwari, Niranjan S. Ghaisas, Kishalay Mitra (2026). **Parameter estimation of the k Turbulence Model for Adverse Pressure Gradient Flows Using Physics-Informed Neural Networks**. In preparation for submission to *Physical Review Fluids.*
- Utkarsh Deep Tiwari, Niranjan S. Ghaisas, Kishalay Mitra (2026). **Influence of yaw misalignment direction and coriolis force on the wake characteristics of a wind turbine**. 14th International Symposium on *Turbulence and Shear Flow Phenomena (TSFP)*, Heidelberg, Germany.
- Utkarsh Deep Tiwari, Kishalay Mitra, Niranjan S. Ghaisas (2025). **Physics-Informed Neural Network Approach for Simulating Flow over an Elliptic Cylinder at Low Reynolds Numbers**. Accepted in the 12th International and 52nd National Conference on *Fluid Mechanics and Fluid Power (FMFP)*, Ahmedabad, India.
- Utkarsh Deep Tiwari, Niranjan S. Ghaisas, Kishalay Mitra (2025). **Evaluating the combined effect of yaw misalignment and wind farm column misalignment on wake losses in finite wind farm arrays using large eddy simulations**. *Physics of Fluids*, Vol. 37 (10). DOI: https://doi.org/10.1063/5.0278295
- Utkarsh Deep Tiwari, Niranjan S. Ghaisas, Kishalay Mitra (2025). **Wind Farm Performance Assessment Using CFD: Key Factors, SOWFA, and Yaw Misalignment Case Study**. In: Mitra, K., Everson, R., Fieldsend, J. (eds) *Optimization, Uncertainty and Machine Learning in Wind Energy Conversion Systems. Engineering Optimization: Methods and Applications. Springer, Singapore*. DOI: https://doi.org/10.1007/978-981-97-7909-3_5
- Utkarsh Deep Tiwari, Niranjan S. Ghaisas, Kishalay Mitra (2023). **Studying the combined effects of yaw angle and wind direction in finite wind farm arrays using large eddy simulations**. 4th *Wind Energy Science Conference (WESC)*, Glasgow, United Kingdom.
- Utkarsh Deep Tiwari, Parmod Kumar (2023). **Investigation of film boiling over tandem arrangement of elliptical cylinders in vertical cross-flow of saturated liquid**. *Applied Thermal Engineering*, Vol. 227. DOI: 10.1016/j.applthermaleng.2023.120308
- Utkarsh Deep Tiwari, Parmod Kumar (2023). **Understanding the Characteristics of Film Boiling Heat Transfer Over Curved Surfaces with Emphasis on Thermal Management of Electronics**. In: Sikarwar, B.S., Sharma, S.K., Jain, A., Singh, K.M. (eds) Advances in Fluid and Thermal Engineering. FLAME 2022. *Lecture Notes in Mechanical Engineering. Springer, Singapore*. DOI: https://doi.org/10.1007/978-981-99-2382-3_56
- Utkarsh Deep Tiwari, Parmod Kumar (2021). **Simulations of boiling heat transfer over the superheated rotating cylindrical surface**. 15th International Conference on *Heat Transfer, Fluid Mechanics and Thermodynamics*, pp. 1994-2000, Amsterdam, Netherlands.

TECHNICAL SKILLS AND RESEARCH EXPERTISE

- **Programming Languages:** C, C++, Fortran, Python
- **Machine Learning Frameworks:** PyTorch, TensorFlow
- **Simulation and Analysis Tools:** OpenFOAM, DeepXDE, Ansys (Meshing), MATLAB, ParaView, Tecplot
- **Specialized Area:** Wind Farm Flows, Atmospheric Boundary Layer Flows, Multiphase Flows, Data-driven and Analytical Modelling of Wind Farms
- **Research Skills:** Computational Fluid Dynamics, Analytical Modelling, Physics-Informed Neural Networks (PINNs) simulations, Scientific Computing

RELEVANT COURSES

- Computational Fluid Dynamics
- Turbulence
- Heat Transfer
- Numerical Methods

AWARDS AND CERTIFICATIONS

Physics-Informed Neural Networks and Applications

KTH Royal Institute of Technology, Stockholm, Sweden

- Gained hands-on experience in PINNs for solving complex differential equations using deep learning.
- Used PINNs architecture to predict the turbulent wake behind a wind turbine.

Introduction to OpenFOAM Development

Skill-Lync

- Developed custom CFD solvers in OpenFOAM using C++, implementing FVM, turbulence models, and pressure-velocity coupling schemes.
- Worked on applied projects involving temperature equation integration, custom function objects, and enhancement of existing solvers.

Prime Minister's Research Fellowship

Ministry of Education, Government of India

- India's highest-funded doctoral fellowship
- Awarded for outstanding academic and research track record

ACADEMIC EVENT MANAGEMENT

- Prime Minister Research Fellowship (PMRF) Annual Symposium, 2025.
- 30th International Conference on Processing of Advanced Materials and Fabrication of Products (PAMFP), 2024.
- Computations of High-Speed, Turbulent, and Complex Flows (CHTC), 2024.
- National Supercomputing Mission Workshop for Training in Computational Fluid Dynamics, 2023.
- 4th International and 19th National Conference on Machines and Mechanisms, 2019.

ADDITIONAL INFORMATION

Languages: Hindi (Fluent), English (Fluent)

Interests: Chess, Hindi Poetry, History, Geo-politics.

REFERENCES

Kishalay Mitra

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Indian Institute of Technology Hyderabad (IIT-H), India

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Relationship: Ph.D. Thesis Supervisor

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Relationship: Ph.D. Thesis Supervisor

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Relationship: M.Tech Thesis Supervisor